Lahore University of Management Sciences

School of Science and Engineering

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Solution Scheme (Prepared by Dr. Hassan Mohy-ud-Din)

Course Title: **AI 500 – Foundations of AI**

Instructor: **Dr. Hassan Mohy-ud-Din**

Type: **Quiz # 1**

Semester: **Fall 2026**

Date: **October 3, 2026**

Marks: **50 points**

Duration: **60 minutes**

Basic Definitions [CLO 1]

Provide a compact response to the following questions.

(a) [3 points] Define **artificial intelligence**. Your definition should capture both the **scientific** and **engineering** character of AI rather than equating AI only with machine learning.

Artificial intelligence is the scientific and engineering discipline concerned with developing computational systems that can receive or perceive information, construct useful representations, reason or learn from evidence, and generate actions or outputs in pursuit of specified objectives under uncertainty and constraints.

Rubric

Recognizes AI as a scientific and engineering discipline	1
Captures intelligent computational capabilities such as representation, reasoning, learning, perception, decision/action	1
Refers to goals/objectives and preferably uncertainty or constraints; does not reduce AI to ML	1

(b) [3 points] Define **temporal leakage** in an AI or machine-learning system. Why does the presence of temporal leakage make an apparently strong evaluation result unreliable?

Temporal leakage occurs when a model uses information that would **not have been available at the time the prediction is supposed to be made**, usually because the feature was measured or recorded later.

It makes evaluation unreliable because the model effectively gains information from the future. Test performance can therefore appear artificially high even though the same information will not be available during real-world prediction.

Rubric

Correctly identifies use of post-index/future information	1.5
Explains that the information would not be available at actual prediction time	0.5
Explains artificially inflated evaluation / failure to reflect deployment performance	1

(c) [3 points] Define **probabilistic calibration**. In one additional sentence, distinguish calibration from **discrimination**.

A probabilistic model is well calibrated when predictions assigned a probability p occur approximately a proportion p of the time. For example, among cases given a predicted risk of 0.8, approximately 80% should experience the outcome.

Discrimination instead concerns how well the model distinguishes or ranks cases that experience the outcome from those that do not.

Rubric

Correct definition of calibration	1.5
Recognizes agreement between predicted probability and observed frequency	0.5
Correct distinction from discrimination/ranking/separation ability	1

Short, Conceptual Questions [CLO 1]

(a) [4 points] Explain why *artificial intelligence is not synonymous with machine learning*. Give **one example of an AI system or technique that does not need to learn from data** and explain briefly why it can still legitimately be regarded as part of AI.

Machine learning is one important approach **within** AI, but AI is broader. AI also includes methods that do not learn parameters from data, such as search, symbolic reasoning, automated planning, constraint satisfaction, optimization, expert systems, and rule-based control.

For example, a shortest-path search algorithm may explore a state space (space of configurations or possibilities) to select a route without learning from historical examples. It is nevertheless part of AI because it performs goal-directed computational reasoning or problem solving.

Rubric

States clearly that ML is a subset/approach within the broader AI field	1.5
Identifies an appropriate non-learning AI method/system	1
Explains why the example still constitutes AI/problem solving/intelligent action	1
Clear conceptual distinction; does not equate automation alone with AI	0.5

(b) [4 points] A model estimates that a patient has a probability of **0.62** of being readmitted to hospital within 30 days. Explain why this number is a **prediction rather than a decision**. Identify **three considerations** that should influence whether an intervention is actually triggered.

The value **0.62** is an estimate of the probability of 30-day readmission. It describes predicted risk; by itself, it specifies **no action**.

To turn it into a decision, the organization must specify a decision rule or policy. Appropriate considerations include any three of:

- relative consequences of false positives and false negatives;
- capacity to provide the intervention;
- expected benefit of intervention;
- calibration/reliability of the probability;
- uncertainty;

- fairness and subgroup consequences;
- availability of human review;
- costs and workflow consequences;
- alternative interventions or thresholds.

Rubric

Explains that 0.62 is an estimated probability/risk, not an action	1
Explains that a decision requires a rule, threshold, or policy linking prediction to action	1
First valid consideration	2/3
Second valid consideration	2/3
Third valid consideration	2/3

(c) [4 points] Two models solve the same prediction task. **Model A** has slightly higher test-set accuracy, whereas **Model B** is simpler, faster, easier to interpret, and considerably cheaper to maintain. Explain why **Model A should not automatically be selected for deployment**. Discuss **any three criteria other than raw predictive accuracy** that should influence the choice.

Higher test-set accuracy does not imply that a model will provide greater overall real-world utility. Deployment requires considering not just predictive performance but also operational, governance, and lifecycle constraints.

Any three of the following are valid:

- Interpretability/explainability: Model B may be easier for users or regulators to understand.
- Latency: Model B may return predictions faster.
- Computational requirements: Model B may require less memory or specialized hardware.
- Maintenance burden: Model B may be easier and cheaper to monitor, update, and retrain.
- Robustness: performance under missing, noisy, or shifted data may differ.
- Calibration: probabilities may be more reliable.
- Governance/auditability: simple models may be easier to document and audit.
- Data requirements: a complex model may require more training data.
- Team expertise: the organization must be able to maintain the deployed system.
- Lifecycle cost: training and maintenance costs matter over the lifetime of the system.

Rubric

Explains why accuracy alone cannot determine deployment choice	1
First well-explained criterion	1
Second well-explained criterion	1
Third well-explained criterion	1
Explains why accuracy alone cannot determine deployment choice	1

Scenario-based Questions [CLO 1]

[21 points] *AI-Assisted Water-Pipe Failure Management:* A city water utility maintains **40,000 underground water pipes**. It wants to use AI to identify pipes at risk of failure so that inspection teams can intervene before a major leak occurs. The available historical data include:

- pipe age and material;

- pipe diameter;
- recent water-pressure sensor readings;
- rainfall measurements;
- previous maintenance records;
- customer complaints about low pressure;
- date and time at which an emergency repair was completed;
- neighborhood;
- recorded pipe failures.

The data science team trains a machine-learning model using a **random 80/20 train-test split** and reports **95% overall accuracy**.

Management proposes the following policy: *For every pipe whose predicted probability of failure exceeds 0.5, automatically shut the corresponding valve and dispatch an inspection team.*

However,

- some pressure sensors report intermittently;
- sensor coverage is much poorer in older neighborhoods;
- the **emergency-repair-completed** field is generally recorded only after a pipe has already failed;
- the city has capacity to inspect only **20 pipes per day**;
- unnecessarily shutting a pipe interrupts water supply to homes and businesses;
- failing to identify an actual imminent failure can cause flooding, infrastructure damage, and substantial repair costs.

Answer the following questions:

(a) [5 points; Problem Formulation] The statement *build an AI model to predict pipe failure* is inadequately formulated. Specify:

- an appropriate **unit of analysis**;
- an **index time**;
- a reasonable **prediction horizon**;
- the **decision-maker**; and
- the **action** that the prediction is intended to support.

There is not necessarily one unique correct formulation. Full credit should be awarded for a **coherent formulation in which all elements are mutually compatible**. A sample response is given below:

- **Unit of analysis:** one individual pipe at a particular prediction/index time.
- **Index time:** for example, the beginning of each day, before inspection assignments are made.
- **Prediction horizon:** probability that the pipe will fail within the next 24 hours, 48 hours, 7 days, etc. Any reasonable horizon receives credit if justified operationally.
- **Decision-maker:** water-utility maintenance/operations manager or inspection-dispatch team.
- **Action:** prioritize/select pipes for inspection, preventive maintenance, intensified monitoring, or (in sufficiently high-risk cases) controlled shutdown.

At the beginning of each day, estimate for every eligible operating pipe the probability of failure during the next seven days using only information available at that time, so that the maintenance team can prioritize the limited daily inspection capacity.

Rubric

Appropriate unit of analysis	1
Clearly defined index time	1
Reasonable prediction horizon	1
Appropriate decision-maker	1
Operationally meaningful action	1

(b) [4 points; Data and representation risks] Identify and explain **two distinct risks** in the available data. At least one of your answers must consider the **emergency-repair-completed** variable. Identify **two additional aspects of evaluation** that should be investigated before deployment.

The emergency-repair-completed timestamp is usually recorded **after failure has already occurred**. If the objective is to predict future failure before intervention, this variable contains information from after the prediction/index time. It therefore creates **temporal leakage** and can cause unrealistically high apparent test performance.

Rubric

identifies temporal leakage: 1
explains why this information is unavailable/inappropriate at prediction time and how it inflates evaluation: 1

Any one well-supported risk may receive the remaining 2 marks.

- Intermittent sensor data / missingness: Pressure measurements are not consistently available. Missingness can degrade predictions and may itself be systematically related to infrastructure condition.
- Unequal sensor coverage / representation bias: Older neighborhoods have poorer sensor coverage, meaning the model observes less information for those areas. Performance may therefore differ systematically across neighborhoods.
- Proxy encoding: Neighborhood could indirectly encode socioeconomic or infrastructure characteristics and create inappropriate dependence on proxy variables.
- Selection/measurement bias: Better-instrumented pipes generate richer observations than poorly instrumented pipes.
- Temporal splitting concern: A random 80/20 split may not test future deployment realistically if patterns, infrastructure, sensors, or operating practices change over time.

Rubric

Correctly identifies a distinct risk	1
Explains its mechanism or consequence	1

(c) [4 points; Evaluation] Explain why the statement *the model achieved 95% accuracy, therefore it is ready for deployment* is inadequate. Identify **two additional aspects of evaluation** that should be investigated before deployment.

Overall accuracy alone is inadequate because it:

- does not distinguish false positives from false negatives;
- may be misleading under class imbalance if pipe failures are rare;
- does not tell whether probabilities are calibrated;
- conceals variation across neighborhoods or infrastructure subgroups;
- says nothing about robustness to missing sensor values;

- says nothing about operational usefulness;
- may have been obtained from a random split that does not match future deployment.

Two additional aspects could include:

- **Sensitivity/recall or false-negative rate**, because missed failures have very high cost.
- **Calibration**, because predicted probabilities are being used to determine intervention.
- **Subgroup performance**, especially across neighborhoods with differing sensor coverage.
- **Robustness**, especially with intermittent/missing sensor data.
- **Temporal validation**, to mimic future deployment.
- **Decision utility/cost**, taking inspection capacity and disruption into account.
- Comparison against an appropriate **baseline**.

Rubric

Explains at least one substantive reason 95% accuracy is insufficient	1
Provides another substantive explanation of why deployment readiness does not follow	1
First appropriate additional evaluation dimension	1
Second appropriate additional evaluation dimension	1

(d) [4 points; From prediction to action] Why is the proposed threshold of **0.5** not automatically appropriate? Explain how both:

- the different consequences of **false positives versus false negatives**, and
- the capacity limit of **20 inspections per day**.

should affect the decision policy.

The threshold 0.5 is not intrinsically optimal. It is simply a numerical cutoff and must be chosen according to the consequences and operational context.

Asymmetric error costs: A false negative may result in flooding, infrastructure damage, and large repair costs. This may justify lowering the threshold to detect more truly vulnerable pipes. However, false positives also matter because unnecessary shutdowns interrupt water supply.

Capacity: The utility can inspect only 20 pipes per day. A threshold that generates 200 daily alerts is operationally useless. The threshold or decision policy should therefore reflect available capacity. For example, ranking pipes by risk and selecting the highest-priority cases subject to the 20-per-day limit.

Optimal Response: ranking plus capacity-constrained prioritization more appropriate.

Rubric

Explains why 0.5 is not a universal/default optimal threshold	1
Correctly discusses asymmetric FP/FN consequences	1
Correctly incorporates the 20-inspection/day capacity constraint	1
Integrates these into a sensible policy: threshold tuning, ranking, triage, human review, etc.	1

(e) [4 points; Deployment and monitoring] State:

- one condition under which the system should abstain or fall back to another procedure, and
- one quantity or behavior that should be continuously monitored after deployment.

Abstention/fallback

The model should abstain or fall back if:

- critical pressure-sensor values are unavailable or clearly invalid;
- incoming data violate expected ranges or schema;
- inputs are sufficiently different from the training distribution;
- confidence/reliability is inadequate;
- the model-serving pipeline fails;
- a pipe belongs to a population on which the model has not been validated.

Fallback could include:

- manual engineering review;
- legacy risk rules;
- intensified monitoring;
- dispatch according to an established non-AI prioritization procedure.

Rubric: 1 mark for the condition + 1 mark for an appropriate response/fallback.

Monitoring

Acceptable monitored quantities include:

- missing sensor rate;
- input/data drift;
- calibration;
- sensitivity or false-negative rate;
- false-positive rate;
- failures missed by the model;
- subgroup performance by neighborhood/infrastructure type;
- alert volume;
- inspection yield;
- latency/uptime;
- proportion of model recommendations overridden;
- real-world pipe failures following deployment.

Rubric: 1 mark for an appropriate monitored quantity + 1 mark for explaining why it matters.

Identifying Biases in an AI System [CLO 1]

[8 points] A company uses an AI system to rank job applicants. The model is trained on ten years of historical hiring data. Applicants from a small set of universities were hired more frequently in the past, and the model consequently assigns them higher scores. The recruitment dashboard prominently highlights the top-ranked applicants, and recruiters rarely inspect candidates ranked lower down. After one year, the model is retrained using the hiring decisions made with assistance from the system.

Identify and briefly explain four distinct sources or forms of bias/problematic feedback present in this scenario.

Rubric: Award **2 points per distinct mechanism**:

- **1 point:** correct identification/name.
- **1 point:** correct explanation tied to the scenario.

Historical bias / inherited institutional bias: Historical hiring decisions favored applicants from particular universities. The training data therefore encode previous institutional preferences. The

model can reproduce these historical patterns rather than objectively estimating applicant suitability.

Proxy bias / proxy encoding: University attended may function as a proxy for factors such as socioeconomic opportunity, geography, access to elite institutions, or historical privilege. The model may therefore indirectly reproduce disparities even without explicitly using protected characteristics.

Caution: This should count only if the student explains the proxy mechanism. Simply repeating "some universities were hired more often" is historical bias rather than automatically a distinct proxy-bias explanation.

Automation bias / interface-induced bias: The dashboard visually highlights top-ranked applicants. Recruiters therefore give disproportionate attention to the model recommendations and rarely independently inspect lower-ranked applicants. Interface design changes human behavior and can cause excessive reliance on automation.

Self-reinforcing feedback loop: Applicants ranked highly receive more human attention and are consequently more likely to be hired. The model is then retrained on these new hiring decisions. Its previous outputs have therefore influenced the data used for future training, reinforcing its original preference.

Other possible answers can fetch full points provided they are clearly distinguished and correctly explained.

Selective exposure: lower-ranked candidates receive little recruiter attention.

Selective labels: meaningful outcomes become observable primarily for candidates who are selected/interviewed/hired; performance of rejected candidates is largely unobserved.

Sampling/representation bias: depending on how the historical dataset is constructed.

Measurement bias: only if the student clearly identifies an imperfect measured variable/label.

Feedback bias: acceptable terminology if the mechanism is accurately explained.

Caution: Avoid double counting. *Recruiters look mostly at highlighted applicants* and *Lower-ranked applicants are rarely reviewed* could both be manifestations of the same **selective exposure/interface mechanism**. A student should not automatically earn 4 points unless the student genuinely distinguishes two different mechanisms.